

CPEN 416 / EECE 5710

Lecture 18: Quantum channels; noise in quantum systems

Thursday 13 November 2025

Announcements

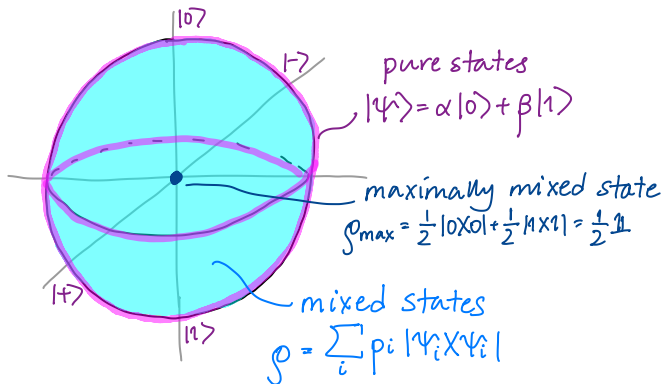
- Quiz 8 on Tuesday (L16-L18)
- Monday's tutorial: observables and expectation values
- Final week schedule updates
 - Mon. 1 Dec: last tutorial, exam review
 - Tues. 2 Dec: quiz 10, groups 01-02 presentations
 - Thurs. 4 Dec: groups 03-05 presentations

Last time

	Pure state	Pure state ρ	Mixed state ρ
States			
Ops.			
Meas.*			

* where $\{\varphi_i\}$ form an orthonormal basis, and $\{P_i\}$ is a set of projectors $P_i = |\varphi_i\rangle\langle\varphi_i|$ or more generally a POVM ($\sum_i P_i = I$).

Last time



Learning outcomes

- 1 Define and apply quantum channels to quantum states
- 2 Express noise in quantum systems using quantum channels
- 3 Compare density matrices with fidelity and trace distance

Quantum channels

To “get inside” the Bloch sphere, we have to apply a different kind of operation: a quantum channel.

A quantum channel Φ *maps* density matrices to density matrices.

Quantum channels are completely positive, trace-preserving (CPTP) linear maps.

- Trace-preserving:
- Positive:
- Completely positive:

Quantum channels

Channels are characterized by a set of **Kraus operators** $\{K_i\}$,

where

A channel's Kraus operators represent, loosely, a set of things that can happen to a system, including *measurement*, and *errors*.

Example: a channel with one Kraus operator is a unitary channel.

Example: measurement

Concrete example: $\{M_m\} = \{\Pi_+, \Pi_-\}$:

Consider mixed state

Intuitively, after measurement expect

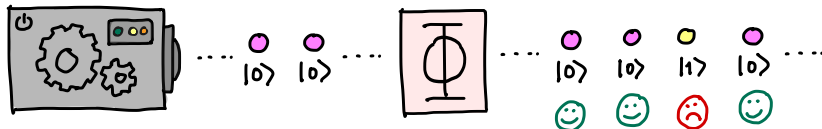
Example: measurement

(Normalized) state after measuring and obtaining outcome m :

Overall state transforms as

The bit flip channel

Suppose a “bit flip” (Pauli X) error occurs with probability p .



How do we write this as a channel? What are the Kraus operators?

The bit flip channel

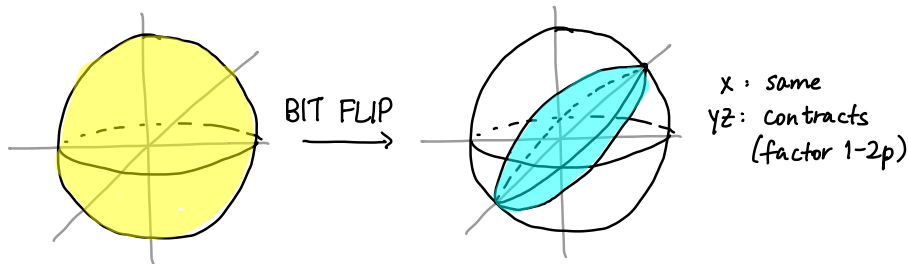
We can visualize the effects of the channel by observing how it deforms the Bloch sphere.

Consider:

- Which states are not affected by a bit flip channel?
- Which states are *most* affected, and where do they go?

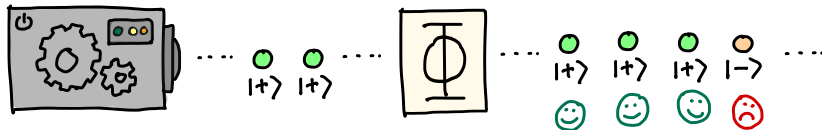
Then look at the density matrices of these states.

The bit flip channel



The phase flip channel

Suppose a “phase flip” (Pauli Z) error occurs with probability p .



How do we write this as a channel? What are the Kraus operators?

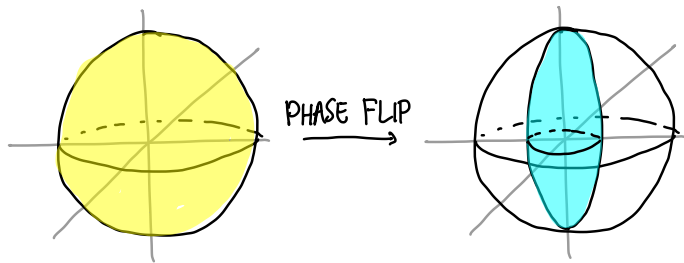
The phase flip channel

Same questions:

- Which states are not affected by a phase flip channel?
- Which states are *most* affected, and where do they go?

Then look at the density matrices of these states.

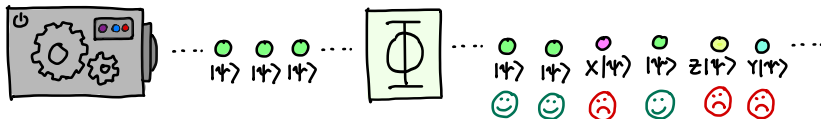
The phase flip channel



z : same
 xy : contracts
(factor $1-2p$)

The depolarizing channel

Suppose each Pauli error occurs with probability $p/3$.



How do we write this as a channel? What are the Kraus operators?

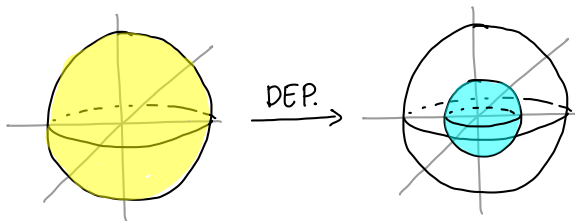
Q: What will happen to the Bloch sphere?

The depolarizing channel

The depolarizing channel

$$\Phi(\rho) = (1 - p)\rho + \frac{p}{3}X\rho X + \frac{p}{3}Y\rho Y + \frac{p}{3}Z\rho Z$$

can also be written as



Amplitude damping channel

Suppose $|1\rangle$ relaxes to $|0\rangle$ with probability p .

How do you think this affects the Bloch sphere?

Comparing density matrices

Exercise: Suppose we prepare a system in the state

$$|\psi\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle$$

If a depolarizing channel with $p = 0.02$ is applied,

- ❶ What's the probability of observing $|0\rangle$?
- ❷ How do we quantify the distance from the original state?

Solution 1: solve by hand. Tedious, but can evaluate

Comparing density matrices

The inner product tells us how close two pure states are:

How close are two mixed states ρ and σ ?

One common metric is the **trace distance**:

Bounded by $0 \leq T(\rho, \sigma) \leq 1$; *lower* trace distance is better.

Comparing density matrices

Can also compare using **fidelity**,

Bounded by $0 \leq F(\rho, \sigma) \leq 1$. *Higher* fidelity is better.

Example: Determine $F(\rho, \sigma)$ if both ρ, σ are pure.

Comparing density matrices

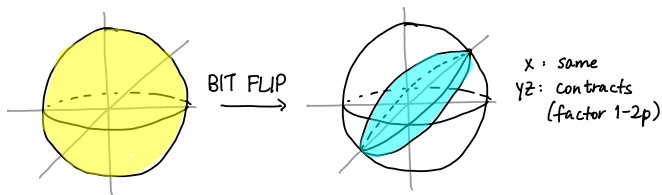
Exercise: Determine $F(\rho, \sigma)$ if ρ is pure but σ is mixed.

Comparing density matrices

Exercise: What is the fidelity of any pure ρ with the maximally mixed state, $\sigma = \frac{1}{2}I$?

Comparing density matrices

Exercise: Determine $F(\rho, \sigma)$ if ρ is pure and $\sigma = \Phi(\rho)$, where Φ is the *bit flip channel* with parameter p .



Next time

Next/ class:

- Intro to quantum error correction

Action items:

- ➊ Prepare for quiz 8
- ➋ Bonus assignment (Piazza)
- ➌ Work on project

Recommended reading:

- From this class: Codebook NT, D; N&C 8.2-8.3, 9.1-9.2
- For next class: Codebook E; N&C 10.1-10.2